Session Objectives

1. Explain **sarcoidosis** and evaluate its etiology and immunology as well as its clinical, microscopic (specifically granulomatous nature), and gross features
2. Outline the **markers** of **activities** and **differential** diagnosis of sarcoidosis
3. Illustrate **pulmonary Langerhans cell histiocytosis** in terms of clinicopathological features, morphological pathological features, and electron microscopic findings
4. Identify **hypersensitivity pneumonitis** in terms of etiology, histopathological findings, and clinical correlations
5. Distinguish **pulmonary eosinophilic syndromes** and their main diagnostic criteria as well as **pulmonary alveolar proteinosis**

Restrictive/Infiltrative/Interstitial Lung Disease features

- ↓ **Diffusing capacity**, ↑ **A-a gradient**
- Normal or mildly increased **FEV1/FVC** ratio &
  - ↓**TLC**, FVC, lung compliance

- Example diagnoses:
  - Sarcoidoses
  - IPF
  - Granulomatosis with polyangiitis
  - Goodpasture syndrome
  - Hypersensitivity pneumonitis
  - Langerhan cell histiocytosis
  - ARDS, neonatal respiratory distress syndrome
  - Pneumoconioses (Occupational Lung diseases)
  - Drug toxicity (ex. bleomycin, busulfan, amiodarone, methotrexate)

Sarcoidosis

- Systemic **non-necrotizing granulomatous** disease
- **Lung** involvement and bilateral hilar lymphadenopathy in 90% of cases
- Disease of disordered immune regulation in genetically predisposed people
  - Age range **20 – 40** years, **female**:male = 2:1
  - Incidence: US whites 10.9/100,000; **African-Americans** 35.5/100,000

Sarcoidosis is a multisystem disorder of unknown cause

- Exposure to antigen?
- Genetic predisposition?
- Unknown cause but cellular mechanisms are known with **evidence of disordered immune regulation** (in genetically susceptible people) resulting from an immunological response to antigenic trigger, such as **noncaseating granulomas**

  - Clinical diagnosis shows multiple clinical presentations: **Dx of exclusion**
    - **Infectious and noninfectious causes** of granulomatous disease must be ruled out (e.g., tuberculosis)

    - Diagnosis established when **clinicoradiological** findings are supported by **histological** evidence of noncaseating (non-necrotizing) epithelioid (plump macrophages) cell granulomas

Sarcoidosis immunologic diagnosis

- **Tuberculin skin test** for cell mediated immunity ↓
- Immunophenotyping of peripheral blood or **BALF** (Bronchoalveolar Lavage Fluid) **cells:**
  - ↑ **CD4/CD8** Ratio
  - ↑ **total cell count** & number of **lymphocytes > 16%**
  - Nearly normal percentage of eosinophils and neutrophils and absence of plasma cells, distinguish most likely diagnosis of sarcoidosis from most common interstitial lung diseases
- **Noncaseating granulomas**

Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2026, Ch. 15

Sarcoidosis - markers of activity

- Recent progressive **dry cough**
- Progressive **dyspnea**
- **Systemic** signs/symptoms: Fatigue, fever, polyarthralgia, erythema nodosum, lymphadenopathy
- Progressive changes on chest **X-rays** or lung CT scans
- **BALF** - CD4+ alveolitis
- Hypercalciuria > hypercalcemia
- **High** serum levels of **ACE**

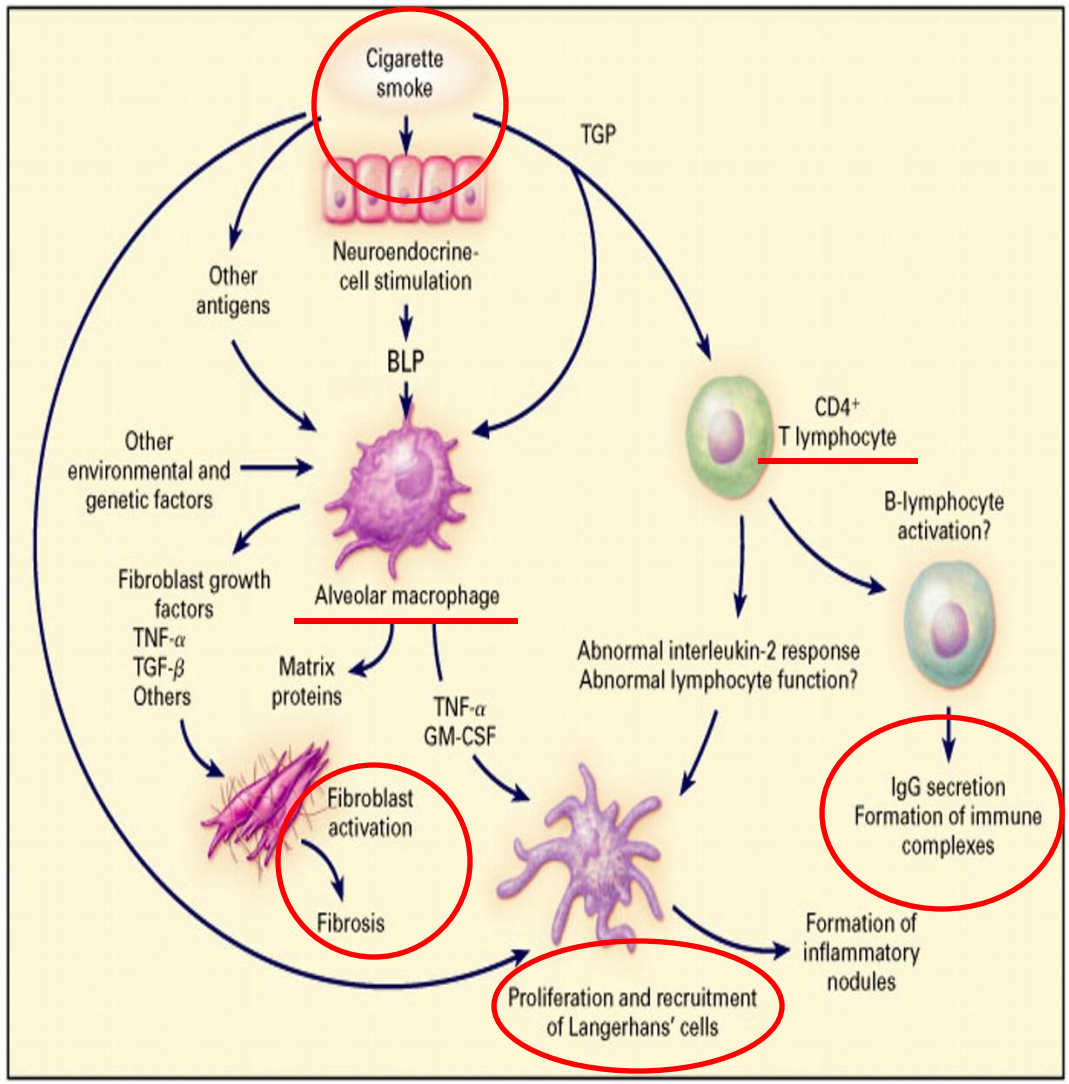
Sarcoidosis markers of activity - ACE (angiotensin-converting enzyme)

- **ACE activity** ↑in sarcoidosis**: High serum** levels in ~ 50-80% of patients
  - Support a diagnosis of sarcoidosis
  - In sarcoidosis, produced by granuloma epithelioid cells & macrophages
- Normal level does not exclude sarcoidosis - ↑ ACE also seen in **other diseases**
- Serum ACE seems to reflects **activity** of disease - ↓ in some patients receiving prednisone

Proposed Pathogenesis of Pulmonary Langerhans Cell Histiocytosis

Primary event: **Cigarette-smoke-induced recruitment** and activation of **Langerhans cells to lung**; potential mechanisms:

- Activate alveolar **macrophages**
- Macrophages produce cytokines → ↑ recruitment **Langerhans** cells activity
- Local expansion of **T** lymphocytes
- Stimulates **B**-lymphocyte → **antibodies** & immune-complex formation
- Fibroblast activation & **fibrosis**



Vassallo R et al. N Engl J Med 2000;342:1969-1978

The NEW ENGLAND JOURNAL of MEDICINE

NEW YORK INSTITUTE OF TECHNOLOGY
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Hypersensitivity pneumonitis

- **A group** of interstitial lung disorders **- immune mediated,** caused by intense, usually long-term exposure to **inhaled organic dusts**
  - Organic dust usually contains antigens made of spores of **bacteria, bacterial products, fungi, animal proteins**
  - Most patients have specific serum **antibodies** against causative agent

- Symptoms typically 4-6 hours after exposure
  - May last ~ **12 hours-days** & **recur** w/ re-exposure

- Examples:
  - **Farmer's** lung (exposure to dusts from humid, warm, newly harvested **hay**, permits proliferation of spores of thermophilic (like warm temperatures) actinomycetes)
  - Pigeon breeder's/ Bird fancier's lung (provoked by proteins from serum, excreta, or feathers of **birds**)
  - Air conditioner/**hot tub** lung (caused by thermophilic **bacteria** in heated water reservoirs)

Hypersensitivity pneumonitis management & histopathology

- Serious **chronic fibrotic lung** disease may be prevented by **removing triggers**

- Histopathologic changes usually centered on bronchioles
  - **Noncaseating granulomas** in 2/3rds of patients, suggesting **T cell-mediated delayed-type** hypersensitivity reaction against antigen

Pulmonary eosinophilic syndromes

- **Rare**, characterized by infiltration of **eosinophils**, recruited by elevated alveolar **IL-5**, & other eosinophil attractants:

  - **Acute eosinophilic pneumonia (AEP) with respiratory failure** – acute, **idiopathic**, rapid onset w/ fever, dyspnea, & respiratory failure
    - Histopathology shows diffuse alveolar damage & **many eosinophils** → responds to **corticosteroids**

  - **Chronic eosinophilic pneumonia (CEP)** – focal cellular consolidation, mostly **peripheral**, aggregates of **lymphocytes & eosinophils** within septal walls & alveoli, may have interstitial fibrosis
    - Signs & symptoms: Cough, fever, night sweats, dyspnea and weight loss → respond to **corticosteroids**
    - **Often idiopathic**

      ↓
  - **Secondary eosinophilia** – occurs in **parasitic**, **fungal**, and **bacterial** infections, hypersensitivity pneumonitis, **drug** allergies, & **eosinophilic granulomatosis w/ polyangiitis** (formerly known as allergic granulomatosis/Churg-Strauss syndrome - believed to be an autoimmune disorder in susceptible patients to an unknown trigger) → **manage underlying condition**

Pulmonary eosinophilic syndromes main diagnostic criteria:

- **Radiographically** identified pulmonary abnormalities need to be **combined** with other diagnostic methods:

- ↑ **peripheral** blood **eosinophilia** (>500 Eos/mm^3)

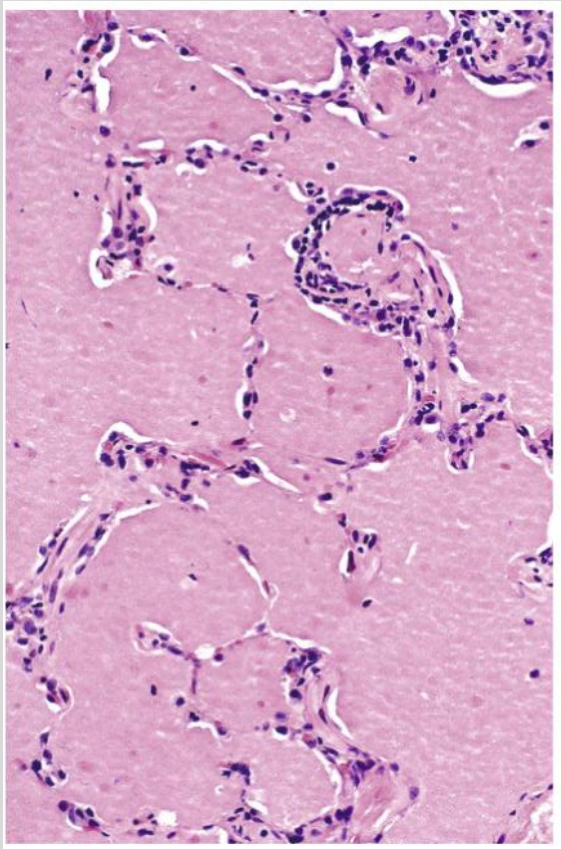
- ↑ eosinophils in **BALF**:
- cut-off of ≥25% for AEP diagnosis
- cut-off of ≥40% for CEP diagnosis

- Lung **tissue eosinophilia** demonstrated in transbronchial or open lung **biopsies**

- **Serology** can help to support diagnoses: CRP, IgE levels, ELISA for coccidioidomycosis, helminths infections for example

Pulmonary alveolar proteinosis (PAP)

- **Rare**
- Caused by deficient **GM-CSF/macrophage dysfunction**
  - **Results** in accumulation of **intra-alveolar precipitates** containing **surfactant proteins**
- **Lamellar bodies (EM)** -↑**normally** made **surfactant**-containing particles
  - Buildup of abnormal surfactant material rich in lamellar bodies that **fill alveoli**
- Precipitate is pink, homogeneous, **PAS+**
- Hereditary – **GM-CSF gene mutations** in neonates affect surfactant proteins
- **Autoimmune,** 90% of patients have antibodies against GM-CSF
- Treated with lung lavage, GM-CSF, secondary cause treatment



H&E stain, lung tissue